Johannes Linder | CV

www.helmholtz-munich.de/icb/johannes-linder

Computer Science PhD with 5+ years of experience working at the intersection of genomics, synthetic biology and deep learning. Junior Group Leader at the Institute of Computational Biology at Helmholtz Munich since 2026. Research focus: Unraveling the regulatory code of transcription, splicing, and post-transcriptional processing using Machine Learning to gain a better understanding of the complicated link between genetic variation and disease, and to enable the rational design of more effective and safer molecular therapies (rAAVs, gene editing, and more).

Education

- **University of Washington** — **Seattle, WA**
  PhD in Computer Science — *2017–2021*

- **KTH Royal Institute of Technology** — **Stockholm, Sweden**
  B.Sc & M.Sc in Computer Science — *2010–2015*

Employment

- **Helmholtz Munich** — **Munich, Germany**
  Junior Group Leader — *March 2026–Current*
  Group Leader at the Institute of Computational Biology. The group focuses on developing accurate and interpretable machine learning models for predicting gene-regulatory processes such as cell type-specific transcription and isoform processing. We aim to use these models to optimize regulatory sequences for multiple therapeutic modalities, e.g. rAAV vectors, and to enable deeper insights into human genetic variation.

- **Calico Life Sciences LLC** — **San Francisco, CA**
  ML Research Scientist — *November 2022–February 2026*
  ML Scientist position in Dr. David Kelley's group, with a focus on interpretable deep learning for regulatory genomics and its application to aging-related disease. Developed long-range sequence-to-function ML models.

- **Stanford University** — **Stanford, CA**
  Postdoctoral Scholar — *September 2021–November 2022*
  Postdoc in Prof. Anshul Kundaje's group at Stanford University. Developed models for predicting the impact of genetic variation on mRNA cleavage and polyadenylation and response to RBP perturbation, and devised methods for engineering cell type-specific promoter (CRISPR) prime edits.

- **University of Washington** — **Seattle, WA**
  Research Assistant — *September 2017–June 2021*
  Research Assistant in Prof. Georg Seelig's group, with a focus on deep learning methods for regulatory genomics, mRNA processing, variant prediction and rational sequence design.

Publications, Proceedings & Presentations

Research Summary. Research focus spanning three inter-connected branches of deep learning for functional genomics: (1) regulatory sequence-to-function learning [J1, J3, J5, J6, J11], (2) model interpretability [J7, J8], and (3) rational sequence design [J2, J4, J9, J10]. Prior work revolves around constructing accurate ML models for predicting gene expression and mRNA isoform abundances, and their response to perturbation, from DNA or RNA sequence. By developing efficient methods for explainability and sequence optimization, these models have enabled researchers to interpret the impact of genetic variation on human health & disease, as well as rationally engineer regulatory elements for molecular therapies.

- Google Scholar Profile: https://scholar.google.com/citations?user=R7IK_EIAAAAJ&hl=en&oi=ao

Peer-reviewed journal articles

J1. Yuan, H., **Linder, J.**, & Kelley, D. R. (2025). Parameter-efficient fine-tuning enables scalable transfer of regulatory sequence models to novel contexts. *Genome Biology*, **27**, 60.
- Link: https://doi.org/10.1186/s13059-026-03935-0

J2. Martyn, G. E., Montgomery, M. T., Jones, H., Guo, K., Doughty, B. R., **Linder, J.**, ... & Engreitz, J. M. (2025). Rewriting regulatory DNA to dissect and reprogram gene expression. *Cell*, **188**, 3349–3366.
- Link: https://doi.org/10.1016/j.cell.2025.03.034

J3. **Linder, J.**, Srivastava, D., Yuan, H., Agarwal, V., & Kelley, D. R. (2025). Predicting RNA-seq coverage from DNA sequence as a unifying model of gene regulation. *Nature Genetics*, **57**, 949–961.
- Link: https://doi.org/10.1038/s41588-024-02053-6

J4. Castillo-Hair, S., Fedak, S., Wang, B., **Linder, J.**, Havens, K., Certo, M., & Seelig, G. (2024). Optimizing $5'$UTRs for mRNA-delivered gene editing using deep learning. *Nature Communications*, **15**, 5284.
- Link: https://doi.org/10.1038/s41467-024-49508-2

J5. Kowalski, M. H., Wessels, H. H., **Linder, J.**, Dalgarno, C., Mascio, I., Choudhary, S., ... & Satija, R. (2024). Multiplexed single-cell characterization of alternative polyadenylation regulators. *Cell*, **187**, 4408–4425.
- Link: https://doi.org/10.1016/j.cell.2024.06.005

J6. Liu, L., Yu, A. M., Wang, X., Soles, L. V., Teng, X., Chen, Y., ..., **Linder, J.**, ... & Shi, Y. (2023). The anticancer compound JTE-607 reveals hidden sequence specificity of the mRNA $3'$ processing machinery. *Nature Structural & Molecular Biology*, **30**, 1947–1957.
- Link: https://doi.org/10.1038/s41594-023-01161-x

J7. **Linder, J.**, Koplik, S. E., Kundaje, A., & Seelig, G. (2022). Deciphering the impact of genetic variation on human polyadenylation using APARENT2. *Genome Biology*, **23**, 232.
- Link: https://doi.org/10.1186/s13059-022-02799-4

J8. **Linder, J.**, La Fleur, A., Chen, Z., Ljubetič, A., Baker, D., Kannan, S., & Seelig, G. (2022). Interpreting neural networks for biological sequences by learning stochastic masks. *Nature Machine Intelligence*, **4**, 41–54.
- Link: https://doi.org/10.1038/s42256-021-00428-6

J9. **Linder, J.**, & Seelig, G. (2021). Fast activation maximization for molecular sequence design. *BMC Bioinformatics*, **22**, 1–20.
- Link: https://doi.org/10.1186/s12859-021-04437-5

J10. **Linder, J.**, Bogard, N., Rosenberg, A. B., & Seelig, G. (2020). A generative neural network for maximizing fitness and diversity of synthetic DNA and protein sequences. *Cell Systems*, **11**, 49–62.
- Link: https://doi.org/10.1016/j.cels.2020.05.007

J11. Bogard, N., **Linder, J.**, Rosenberg, A. B., & Seelig, G. (2019). A deep neural network for predicting and engineering alternative polyadenylation. *Cell*, **178**, 91–106.
- Link: https://doi.org/10.1016/j.cell.2019.04.046

Conference proceedings

C1. **Linder, J.**, Chen, Y-J., Wong, D., Seelig, G., Ceze, L., & Strauss, K. (2021, September). Robust digital molecular design of binarized neural networks [Paper presentation]. *DNA* 27, Virtual Conference.
- Link: https://doi.org/10.4230/LIPIcs.DNA.27.1

C2. **Linder, J.**, La Fleur, A., Chen, Z., Ljubetic, A., Baker, D., Kannan, S., & Seelig, G. (2020, November). Efficient inference of nonlinear feature attributions with Scrambling Neural Networks [Spotlight presentation]. *MLCB* 2020, Virtual Conference.
- Link: https://drive.google.com/file/d/142tmyEMLUSsV-IEkN-NFcEUd7-LFwaAF/view?usp=sharing

C3. **Linder, J.**, Bogard, N., Rosenberg, A. B., & Seelig, G. (2019, December). Deep exploration networks for rapid engineering of functional DNA sequences [Paper presentation]. *MLCB* 2019, Vancouver, Canada.
- Link: https://github.com/johli/genesis/blob/master/mlcb_exploration_nets.pdf?raw=true

Preprints

P1. Chao, K-H., Magzoub, M. M., Stoops, E., Hackett, S., **Linder, J.**, & Kelley, D. R. (2025). Predicting dynamic expression patterns in budding yeast with a fungal DNA language model. *bioRxiv*.
- Link: https://www.biorxiv.org/content/10.1101/2025.09.19.677475v1

P2. **Linder, J.**, Yuan, H., & Kelley, D. R. (2025). Predicting cell type-specific coverage profiles from DNA sequence. *bioRxiv*.
- Link: https://www.biorxiv.org/content/10.1101/2025.06.10.658961v1

P3. Holmes, I., **Linder, J.**, & Kelley, D. R. (2025). Selective State Space Models Outperform Transformers at Predicting RNA-Seq Read Coverage. *bioRxiv*.
- Link: https://www.biorxiv.org/content/10.1101/2025.02.13.638190v1

P4. Koplik, S. E., Angela, M. Y., Shelby, M. R., Fonseca, G. C., Roco, C. M., Zhang, Y., ..., **Linder, J.**, & Seelig, G. (2025). Massively parallel assay of human splice variants reveals cis-regulatory drivers of disease-associated and cell type-specific splicing regulation. *bioRxiv*.
- Link: https://www.biorxiv.org/content/10.1101/2025.10.12.681955v1

Presentations and posters

O1. **Linder, J.** (2025, September). Predicting Gene-Regulatory Function from DNA Sequence with Deep Learning. Presented at the Variant to Function (V2F) Symposium 2025, Cambridge, MA.

O2. **Linder, J.**, Fish, L., Karner, H., Goodarzi, H., & Kelley, D. R. (2025, September). Joint prediction of mRNA expression, degradation rates and RBP binding events. Presented at the 7th conference on Machine Learning for Computational Biology (MLCB 2025), New York, NY.

O3. **Linder, J.**, Kundaje, A., & Seelig, G. (2022, May). Interpreting Cis-Regulatory Polyadenylation Variants with Residual Neural Networks and Attention Masks. Presented at CSHL Biology of Genomes 2022 (BoG 2022), Cold Spring Harbor, NY.
- Associated with journal article [J7].

O4. **Linder, J.**, Chen, Y-J., Wong, D., Seelig, G., Ceze, L., and Strauss, K. (2021, September). Robust Digital Molecular Design of Binarized Neural Networks. Presented at the 27th International Conference on DNA Computing and Molecular Programming (DNA 27), Virtual Conference.
- Associated with conference proceeding [C1].

O5. **Linder, J.**, La Fleur, A., Chen, Z., Ljubetic, A., Baker, D., Kannan, S., & Seelig G. (2020, November). Efficient inference of nonlinear feature attributions with Scrambling Neural Networks. Spotlight presentation and poster at the 2nd conference on Machine Learning for Computational Biology (MLCB 2020), Virtual Conference.
- Associated with conference proceeding [C2].

O6. **Linder, J.**, Bogard, N., Rosenberg, A. B., & Seelig, G. (2019, December). Deep exploration networks for rapid engineering of functional DNA sequences. Presented at the 1st conference on Machine Learning for Computational Biology (MLCB 2019), Vancouver, Canada. Travel award recipient from Recursion.
- Associated with conference proceeding [C3].

O7. **Linder, J.**, Wong, D., Chen, Y-J., Ceze, L., Seelig, G., & Strauss, K. (2019, August). Binarized Neural Networks as Droplet-Mediated Strand Displacement Cascades. Poster at the 25th International Conference on DNA Computing and Molecular Programming (DNA 25), Seattle, Washington.

O8. Bogard, N., **Linder, J.**, Rosenberg, A. B., & Seelig, G. (2019, July). A Deep Neural Network for Predicting and Engineering Alternative Polyadenylation. Poster at the ENCODE 2019 Users Meeting, Seattle, Washington.
- Associated with journal article [J11].

O9. Bogard, N., **Linder, J.**, Rosenberg, A. B., & Seelig, G. (2018, October). Predicting the impact of cis-regulatory variation on alternative polyadenylation; (Program #326). Abstract for the Annual Meeting of The American Society of Human Genetics, San Diego, California. www.ashg.org/2018meeting.
- Associated with journal article [J11].

Community Involvement

- **Kipoi**
  Scientific Committee — *December 2021–Current*
  Scientific committee member for the Kipoi seminar series (https://kipoi.org/seminar/). The series, which is often attended by > 100 people, focuses on deep learning for regulatory genomics and synthetic biology.

Awards and Fellowships

- **The Sweden-America Foundation**
  Research fellowship — *June 2018–June 2019*